

Task 1

1. Adding a disk to VM

To add a new disk to VM, we need to right click it in VMWare Workstation Pro and select Settings. There we add new hard disk. I prefer HDD.

Then we boot the system and run `lsblk` command to see disks:

```
[root@rhel-9 ~ #] lsblk
NAME            MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda              8:0    0   2G  0 disk
sr0             11:0    1 12.7G  0 rom  /mnt
nvme0n1         259:0    0   10G  0 disk
├─nvme0n1p1     259:1    0 600M  0 part /boot/efi
├─nvme0n1p2     259:2    0    1G  0 part /boot
├─nvme0n1p3     259:3    0  8.4G  0 part
└─rhel-root    253:0    0  8.4G  0 lvm  /
```

2.Stratis Packages

First, we need to download the stratis service and command line utilities:

```
dnf install stratisd stratis-cli
```

Then we need to start and enable its service to launch at boot:

```
[root@rhel-9 ~ #] systemctl enable --now stratisd
[root@rhel-9 ~ #] systemctl status stratisd
● stratisd.service - Stratis daemon
   Loaded: loaded (/usr/lib/systemd/system/stratisd.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-04-02 17:13:52 +04; 40s ago
     Docs: man:stratisd(8)
    Main PID: 9273 (stratisd)
      Tasks: 6 (limit: 10424)
    Memory: 2.3M (peak: 2.7M)
       CPU: 30ms
    CGroup: /system.slice/stratisd.service
            └─9273 /usr/libexec/stratisd --log-level debug

Apr 02 17:13:52 rhel-9.7 systemd[1]: Starting Stratis daemon...
Apr 02 17:13:52 rhel-9.7 stratisd[9273]: [2026-04-02T13:13:52Z INFO stratisd::stratis::run] stratis daemon version 3.
Apr 02 17:13:52 rhel-9.7 stratisd[9273]: [2026-04-02T13:13:52Z INFO stratisd::stratis::run] Using StratEngine
Apr 02 17:13:52 rhel-9.7 stratisd[9273]: [2026-04-02T13:13:52Z INFO stratisd::engine::strat_engine::liminal::identify
Apr 02 17:13:52 rhel-9.7 stratisd[9273]: [2026-04-02T13:13:52Z INFO stratisd::stratis::ipc_support::dbus_support] D-B
Apr 02 17:13:52 rhel-9.7 systemd[1]: Started Stratis daemon.
```

3.Creating Pool

We need to remove all block devices where we want to create stratis.

```
[root@rhel-9 ~ #] wipefs --all /dev/sda
```

Then we create a pool named "idtechpool":

```
[root@rhel-9 ~ #] stratis pool create idtechpool /dev/sda
```

Now we can verify the pool is created successfully:

```
[root@rhel-9 ~ #] stratis pool list
```

Name	Total / Used / Free	Properties	UUID	Alerts
idtechpool	2 GiB / 524 MiB / 1.49 GiB	~Ca,~Cr, 0p	8e3303e9-1578-40c9-aa4b-4a7f43c41782	WS001

4.Creating File systems

We create and check our new filesystems (idtech1 and idtech2):

```
[root@rhel-9 ~ #] stratis filesystem create --size 600MiB idtechpool idtech1
[root@rhel-9 ~ #] stratis filesystem create --size 700MiB idtechpool idtech2
[root@rhel-9 ~ #] stratis fs list idtechpool
```

Pool	Filesystem	Total / Used / Free / Limit	Device	UUID
idtechpool	idtech1	600 MiB / 69 MiB / 531 MiB / None	/dev/stratis/idtechpool/idtech1	178a426e-b489-40b4-bd35-4dcf903abaa9
idtechpool	idtech2	700 MiB / 69 MiB / 631 MiB / None	/dev/stratis/idtechpool/idtech2	7048d242-9edf-4be9-8b02-bebd7a935382

The first filesystem is 600 MiB, the second is 700 MiB.

5.Adding a new disk

Now it is time to add a new disk to our pool. First, we need to add a new disk from the VMWare Workstation Pro. We need to right click VM and add hard disk here. I defined its capacity 3 GB.

Let's see it using `lsblk`:

```
[root@rhel-9 ~ #] lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	2G	0	disk	
├─stratis-1-private-ae38564bdd874e31ba58b587ef666e47-physical-originsub	253:1	0	2G	0	stratis	
│ └─stratis-1-private-ae38564bdd874e31ba58b587ef666e47-flex-thinmeta	253:2	0	4M	0	stratis	
│ │ └─stratis-1-private-ae38564bdd874e31ba58b587ef666e47-thinpool-pool	253:5	0	1.5G	0	stratis	
│ │ │ └─stratis-1-ae38564bdd874e31ba58b587ef666e47-thin-fs-d25b20b921754a3f9a4cc6404949c1c1	253:6	0	1T	0	stratis	
│ │ │ └─stratis-1-ae38564bdd874e31ba58b587ef666e47-thin-fs-d39640cd890d4d47bd34de09816efafa	253:7	0	1T	0	stratis	
│ │ └─stratis-1-private-ae38564bdd874e31ba58b587ef666e47-flex-thindata	253:3	0	1.5G	0	stratis	
│ │ └─stratis-1-private-ae38564bdd874e31ba58b587ef666e47-thinpool-pool	253:5	0	1.5G	0	stratis	
│ │ │ └─stratis-1-ae38564bdd874e31ba58b587ef666e47-thin-fs-d25b20b921754a3f9a4cc6404949c1c1	253:6	0	1T	0	stratis	
│ │ │ └─stratis-1-ae38564bdd874e31ba58b587ef666e47-thin-fs-d39640cd890d4d47bd34de09816efafa	253:7	0	1T	0	stratis	
└─stratis-1-private-ae38564bdd874e31ba58b587ef666e47-flex-mdv	253:4	0	512M	0	stratis	
sdb	8:16	0	3G	0	disk	

As you can see, its name is `sdb`.

We can use `wipefs --all /dev/sdb` to remove existing stuff in this disk, but that is not mandatory, because we newly inserted this disk (there is nothing on it yet)

We use this command to add a new disk to pool:

```
[root@rhel-9 ~ #] stratis pool add-data idtechpool /dev/sdb
[root@rhel-9 ~ #] stratis pool list
```

Name	Total / Used / Free	Properties	UUID	Alerts
idtechpool	5 GiB / 1.58 GiB / 3.42 GiB	~Ca,~Cr, 0p	ae38564b-dd87-4e31-ba58-b587ef666e47	

As you can see the size of the pool has increased.

6.Extending Pool

Stratis filesystems don't work like LVMs — **we cannot resize them manually**, Instead, Stratis uses thin provisioning, that means the filesystems can grow automatically from the pool as needed. The "size" we set at creation is just an initial allocation hint, not a hard limit.

The closest thing to that is might be LIMIT size. So, we can define it 1 GiB, so even with automatic increase it cannot pass 1 GiB.

```
[root@rhel-9 ~ #] stratis fs set-size-limit idtechpool idtech1 1GiB
[root@rhel-9 ~ #] stratis fs list idtechpool
```

Pool	Filesystem	Total / Used / Free / Limit	Device	UUID
idtechpool	idtech1	600 MiB / 69 MiB / 531 MiB / 1 GiB	/dev/stratis/idtechpool/idtech1	178a426e-b489-40b4-bd35-4dcf903abaa9
idtechpool	idtech2	700 MiB / 69 MiB / 631 MiB / None	/dev/stratis/idtechpool/idtech2	7048d242-9edf-4be9-8b02-bebd7a935382

However, there is a trick which we can use to set its real size 1GiB. To achieve that, we need to define its limit size as 1 GiB as I did above. Then we need to mount it. We create a file inside the mountpoint. Then we use this command to fill this file:

```
`dd if=/dev/zero of=/str-dir1/file`
```

```
[root@rhel-9 ~ #] stratis fs list
```

Pool	Filesystem	Total / Used / Free / Limit	Device	UUID
idtechpool	idtech1	600 MiB / 69 MiB / 531 MiB / 1 GiB	/dev/stratis/idtechpool/idtech1	22537a03-0fd3-403c-8670-d6ff706df0f1
idtechpool	idtech2	550 MiB / 69 MiB / 481 MiB / None	/dev/stratis/idtechpool/idtech2	9bd5da33-52ff-4f92-95e8-abcb8a85691a

```
[root@rhel-9 ~ #] cd /str-dir1
[root@rhel-9 /str-dir1 #] touch file
[root@rhel-9 /str-dir1 #] dd if=/dev/zero of=./file
dd: writing to './file': No space left on device
1028073+0 records in
1028072+0 records out
526372864 bytes (526 MB, 502 MiB) copied, 3.65935 s, 144 MB/s
[root@rhel-9 /str-dir1 #] stratis fs list
```

Pool	Filesystem	Total / Used / Free / Limit	Device	UUID
idtechpool	idtech1	1 GiB / 572 MiB / 452 MiB / 1 GiB	/dev/stratis/idtechpool/idtech1	22537a03-0fd3-403c-8670-d6ff706df0f1
idtechpool	idtech2	550 MiB / 69 MiB / 481 MiB / None	/dev/stratis/idtechpool/idtech2	9bd5da33-52ff-4f92-95e8-abcb8a85691a

(I have mounted `idtech1` filesystem to `/str-dir1` directory, I will talk about in the next part of the task)

7.Mounting and Checking Filesystems

To mount the filesystems to directories, first I have created directories and then added lines to /etc/fstab.

```
[root@rhel-9 ~ #] mkdir /str-dir1 /str-dir2
```

I have added it to /etc/fstab:

```
[root@rhel-9 ~ #] tail -n2 /etc/fstab
/dev/stratis/idtechpool/idtech1 /str-dir1 xfs defaults,x-systemd.requires=stratis-fstab-setup@pool-uuid.service,x-systemd.after=stratis-fstab-setup@pool-uid.service 0 0
/dev/stratis/idtechpool/idtech2 /str-dir2 xfs defaults,x-systemd.requires=stratis-fstab-setup@pool-uuid.service,x-systemd.after=stratis-fstab-setup@pool-uid.service 0 0
```

After rebooting, it worked properly.

Task 2

1.Adding New Disk and Create a Partition

To add a new disk, we need to right click to our VM and add Hard Disk from here.

Then we can check it inside VM:

```
[root@rhel-9 ~ #] lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0    6G  0 disk
sr0          11:0    1 12.7G  0 rom  /mnt
nvme0n1      259:0    0   10G  0 disk
├─nvme0n1p1   259:1    0 600M  0 part /boot/efi
├─nvme0n1p2   259:2    0    1G  0 part /boot
├─nvme0n1p3   259:3    0   8.4G  0 part
└─rhel-root 253:0    0   8.4G  0 lvm  /
```

Then we need to create partitions from the disk, and I create one partition from the disk. I have used MBR to divide the disk:

```
[root@rhel-9 ~ #] fdisk /dev/sda

Welcome to fdisk (util-linux 2.37.4).
Changes will remain in memory only, until you decide to write them.
Be careful before using the write command.

Device does not contain a recognized partition table.
Created a new DOS disklabel with disk identifier 0x5f0232f8.

Command (m for help): n
Partition type
  p   primary (0 primary, 0 extended, 4 free)
  e   extended (container for logical partitions)
Select (default p):

Using default response p.
Partition number (1-4, default 1):
First sector (2048-12582911, default 2048):
Last sector, +/-sectors or +/-size{K,M,G,T,P} (2048-12582911, default 12582911):

Created a new partition 1 of type 'Linux' and of size 6 GiB.

Command (m for help): w
The partition table has been altered.
Calling ioctl() to re-read partition table.
Syncing disks.
```

We can see it:

```
[root@rhel-9 ~ #] lsblk
NAME        MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda           8:0    0    6G  0 disk
└─sda1        8:1    0    6G  0 part
```

2.Create VG and LV

Then we need to create a VG from that partition:

```
[root@rhel-9 ~ #] vgcreate vgvd0 /dev/sda1
WARNING: adding device /dev/sda1 with idname /dev/sda1 which
Physical volume "/dev/sda1" successfully created.
Volume group "vgvd0" successfully created
[root@rhel-9 ~ #] vgs #To check whether it is created or not
VG      #PV #LV #SN Attr   VSize  VFree
rhel    1   1   0 wz--n-   8.41g    0
vgvd0   1   0   0 wz--n-  <6.00g <6.00g
```

Then we create LV which has VDO:

```
[root@rhel-9 ~ #] lvcreate -n lvm-vdo --type vdo --size 5G --virtualsize 50G vgvd0
The VDO volume can address 2 GB in 1 data slab.
It can grow to address at most 16 TB of physical storage in 8192 slabs.
If a larger maximum size might be needed, use bigger slabs.
Logical volume "lvm-vdo" created.
[root@rhel-9 ~ #] lvs
LV      VG      Attr   LSize  Pool   Origin Data%  Meta%   Move Log Cpy%Sync Convert
root    rhel    -wi-ao--- 8.41g
lvm-vdo vgvd0   vwi-a-v--- 50.00g vpool0      0.00
vpool0  vgvd0   dwi----- 5.00g      60.03
```

3.Mounting to Fstab

Then we need to mount our LV. To mount permanently we need to add a line to /etc/fstab. First, I take its UUID with `blkid`:

```
[root@rhel-9 ~ #] blkid /dev/vgvd0/lvm-vdo
/dev/vgvd0/lvm-vdo: UUID="7b5d80a3-47f7-4830-83f7-af0ca251c051" TYPE="xfs"
```

Then create the directory where we want to mount:

```
[root@rhel-9 ~ #] mkdir /vdo-dir
```

Then we add this to fstab file:

```
[root@rhel-9 ~ #] tail -n1 /etc/fstab
UUID=7b5d80a3-47f7-4830-83f7-af0ca251c051 /vdo-dir xfs defaults 0 0
```

To ensure everything works perfectly, we need to reboot the system.

```
[root@rhel-9 ~ #] lsblk
```

NAME	MAJ:MIN	RM	SIZE	RO	TYPE	MOUNTPOINTS
sda	8:0	0	6G	0	disk	
└─sda1	8:1	0	6G	0	part	
└─vgvdo-vpool0_vdata	253:1	0	5G	0	lvm	
└─vgvdo-vpool0-vpool	253:2	0	50G	0	lvm	
└─vgvdo-lvm--vdo	253:3	0	50G	0	lvm	/vdo-dir



```
[root@rhel-9 ~ #] df -h
```

Filesystem	Size	Used	Avail	Use%	Mounted on
devtmpfs	4.0M	0	4.0M	0%	/dev
tmpfs	838M	0	838M	0%	/dev/shm
tmpfs	335M	6.3M	329M	2%	/run
efivarfs	256K	57K	195K	23%	/sys/firmware/efi
/dev/mapper/rhel-root	8.4G	2.4G	6.0G	29%	/
/dev/nvme0n1p2	960M	243M	718M	26%	/boot
/dev/nvme0n1p1	599M	7.4M	592M	2%	/boot/efi
/dev/sr0	13G	13G	0	100%	/mnt
/dev/mapper/vgvdo-lvm--vdo	50G	389M	50G	1%	/vdo-dir

